Notes on Investment and Certainty Equivalents

Optimal Investment under Expected Utility

Setup

Consider an agent with initial wealth $w > 0$ who allocates resources between two assets.

The first asset is risk-free: each euro invested yields $1 + r$, where $r > 0$.

The second asset is risky: each euro invested yields a random gross return $\theta \in \Theta$, where θ is a real-valued random variable with cumulative distribution function F.

If the agent invests x euros in the risky asset (allowing $x > w$, i.e., borrowing at the risk-free rate), final wealth is

$$f(x, \theta) \;=\; x\theta + (1 + r)(w - x) \;=\; x(\theta - (1 + r)) + w(1 + r).$$

Expected utility is

$$U(x) \;=\; \int_{\Theta} u(f(x, \theta))\, dF(\theta) \;=\; \mathbb{E}\big[u(f(x, \theta))\big],$$

where u is strictly increasing, concave, and twice continuously differentiable. The choice problem is

$$x^* \in \arg\max_{x \geq 0} U(x).$$

Concavity of the objective

Let $x = dx_1 + (1 - d)x_2$, with $d \in [0, 1]$. Since $f(\cdot, \theta)$ is affine in its first argument,

$$f(x, \theta) \;=\; f(dx_1 + (1 - d)x_2, \theta) \;=\; df(x_1, \theta) + (1 - d)f(x_2, \theta).$$

By concavity of u,

$$u(f(x, \theta)) \;\geq\; d\, u(f(x_1, \theta)) + (1 - d)\, u(f(x_2, \theta)).$$

Integrating both sides with respect to F,

$$U(x) \;\geq\; dU(x_1) + (1 - d)U(x_2).$$

Hence U is concave in x.

First-order analysis

Differentiate under the integral sign:

$$U'(x) = \int_{\Theta} u'(f(x, \theta))\, f_x(x, \theta)\, dF(\theta).$$

Because

$$f_x(x, \theta) = \theta - (1 + r),$$

we obtain

$$U'(x) = \int_\Theta u'(f(x,\theta))(\theta - (1+r))\, dF(\theta) = \mathbb{E}\big[u'(f(x,\theta))(\theta - (1+r))\big].$$

Evaluating at $x = 0$, note that $f(0,\theta) = w(1+r)$. Thus

$$U'(0) = \mathbb{E}\big[u'(w(1+r))(\theta - (1+r))\big] = u'(w(1+r))\big(\mathbb{E}[\theta] - (1+r)\big).$$

Implications for the optimal investment

Case $\mathbb{E}[\theta] < 1 + r$

Then $U'(0) < 0$. Since U is concave, U' is nonincreasing, so $U'(x) \le U'(0) < 0$ for all $x \ge 0$. Therefore the maximizer over $x \ge 0$ is

$$x^* = 0.$$

Case $\mathbb{E}[\theta] = 1 + r$

Then $U'(0) = 0$. Concavity implies $U'(x) \le 0$ for all $x \ge 0$. Hence $x^* = 0$ is still optimal.

Case $\mathbb{E}[\theta] > 1 + r$

Then $U'(0) > 0$. Hence $x = 0$ cannot be optimal. If a maximizer exists, it must satisfy

$$x^* > 0.$$

Certainty Equivalents and Decreasing Relative Risk Aversion in Multiplicative Lotteries

Setup: multiplicative risk and certainty equivalents

Consider an agent with initial wealth $w > 0$ who evaluates *multiplicative* lotteries over $\mathbb{R}_+$. A lottery is represented by a nonnegative random variable Y with cdf F. Preferences are represented by

$$U(w; F) \;=\; \int_{\mathbb{R}_+} u(wy)\, dF(y),$$

where $u : \mathbb{R}_+ \to \mathbb{R}$ is twice differentiable, increasing, and concave.

Define the certainty–equivalent final wealth $\bar{c}(w; F, u)$ by

$$u(\bar{c}(w; F, u)) \;=\; \int_{\mathbb{R}_+} u(wy)\, dF(y),$$

and the certainty–equivalent gross return by

$$CE(w; F, u) \;=\; \frac{\bar{c}(w; F, u)}{w}.$$

From scaled utility to relative risk aversion

Fix $w > 0$ and define the scaled utility (utility over gross returns)

$$u_w(y) \;=\; u(wy).$$

This is the utility function the agent implicitly uses when she treats y as the "outcome" (a multiplicative factor).

The certainty equivalent condition can be written as

$$u(\bar{c}(w; F, u)) \;=\; \int u(wy)\, dF(y).$$

Since $CE(w; F, u) = \bar{c}(w; F, u)/w$, we can solve for $\bar{c}(w; F, u)$ as

$$\bar{c}(w; F, u) \;=\; w\, CE(w; F, u).$$

Substituting into the certainty equivalent definition yields

$$u\big(w\, CE(w; F, u)\big) \;=\; \int u(wy)\, dF(y).$$

Now compute derivatives of u_w. First derivative:

$$u_w'(y) \;=\; \frac{d}{dy} u(wy) \;=\; u'(wy) \cdot w \;=\; w\, u'(wy).$$

Second derivative:

$$u_w''(y) \;=\; \frac{d}{dy}\big(w\, u'(wy)\big) \;=\; w \cdot u''(wy) \cdot w \;=\; w^2 u''(wy).$$

The Arrow–Pratt coefficient of *absolute* risk aversion for u_w at outcome y is

$$r_A(y; u_w) \;=\; -\frac{u_w''(y)}{u_w'(y)} \;=\; -\frac{w^2 u''(wy)}{w u'(wy)} \;=\; -\frac{w\, u''(wy)}{u'(wy)}.$$

Evaluating at $y = 1$,

$$r_A(1; u_w) \;=\; -\frac{w\, u''(w)}{u'(w)}.$$

But the Arrow–Pratt coefficient of *relative* risk aversion for u is

$$R(x) \;=\; -\frac{x u''(x)}{u'(x)}.$$

Setting $x = w$, we obtain

$$R(w) \;=\; -\frac{w u''(w)}{u'(w)} \;=\; r_A(1; u_w).$$

So, in a multiplicative setting, relative risk aversion of u at wealth w coincides with the absolute risk aversion of the scaled utility u_w evaluated at the reference point $y = 1$.

Monotonicity link: DRRA and certainty–equivalent returns

For any fixed distribution F, define

$$\bar{c}(w) \equiv \bar{c}(w; F, u), \qquad CE(w) \equiv CE(w; F, u) = \frac{\bar{c}(w)}{w}.$$

The quantity featured in the statement,

$$\frac{w}{\bar{c}(w)},$$

can be rewritten using $CE(w)$:

$$\frac{w}{\bar{c}(w)} = \frac{w}{w\,CE(w)} = \frac{1}{CE(w)}.$$

Therefore,

$$\frac{w}{\bar{c}(w)} \text{ is decreasing in } w \quad \Longleftrightarrow \quad \frac{1}{CE(w)} \text{ is decreasing in } w \quad \Longleftrightarrow \quad CE(w) \text{ is increasing in } w.$$

So the two monotonicity statements are exactly the same.

The key economic content is that decreasing relative risk aversion (DRRA),

$$R(x) = -\frac{xu''(x)}{u'(x)} \quad \text{decreasing in } x,$$

means that as wealth increases the agent becomes *less* averse to proportional (multiplicative) risks. In this multiplicative environment, "wealth going up" corresponds to comparing the scaled utilities $u_w(\cdot) = u(w\cdot)$ across different values of w. DRRA implies that higher w leads to a lower effective level of aversion to the same distribution of gross returns Y, which is captured by a higher certainty–equivalent gross return $CE(w)$.

Economic interpretation

As wealth increases, the agent becomes less averse to proportional risk. As a result, the certainty–equivalent return increases with wealth. Equivalently, richer agents require a smaller compensation *relative to their wealth* to accept the same multiplicative risky lottery.